Windows Endpoint Threat Hunting with Sysmon

Defensive Monitoring Lab Report

Project type: Defensive endpoint monitoring and threat-hunting lab

Platform: Windows (AMD64), Windows PowerShell 5.1

Telemetry source: Microsoft Sysmon 15.22

Status: Initial lab completed; findings are based on controlled benign activity and a review of recent process events.

Executive summary

This project established a basic Windows endpoint monitoring capability using Microsoft Sysmon. The Sysmon service was verified as running, and the Microsoft-Windows-Sysmon/Operational event log was confirmed enabled. Sysmon Event ID 1 (Process Create) was queried to examine process execution, command-line details, and parent-child process relationships.

Controlled tests captured Notepad and PowerShell activity. A separate PowerShell process ran the harmless Get-Date command. A review of recent process events also showed Microsoft Edge, Zoom, and Windows components such as dllhost.exe, conhost.exe, and sdbinst.exe.

No malicious activity was established by these exercises. This was an initial telemetry validation and baseline review, not a full incident investigation or proof that the endpoint is free of threats.

Objectives

- Install and verify Sysmon.
- Confirm the Sysmon Operational event log is available and enabled.
- Retrieve Event ID 1 process-creation records.
- Examine process paths, command lines, process IDs, parent processes, user context, integrity level, and available hashes.
- Separate expected test activity from activity requiring further investigation.
- Document evidence and identify next steps.

Lab environment and tools

- Windows endpoint with AMD64 architecture.
- Windows PowerShell 5.1.
- Microsoft Sysmon v15.22, installed from C:\Tools\Sysmon\Sysmon64.exe.
- Windows Event Log: Microsoft-Windows-Sysmon/Operational.

- Notepad and PowerShell used for controlled benign process tests.

No malware was executed. The test commands created ordinary process activity for telemetry validation.

Method

1. Confirmed AMD64 architecture and the Sysmon 64-bit executable.
2. Installed Sysmon.
3. Verified the Sysmon64 service was running.
4. Confirmed the Sysmon Operational event log was enabled.
5. Queried Event ID 1 process-creation records.
6. Launched Notepad with a distinctive test argument and found the resulting event.
7. Launched a separate PowerShell process to execute Get-Date and queried the log for the command line.
8. Reviewed recent process events.

Findings

Finding 1 — Sysmon installation and logging

The service query returned Running for Sysmon64. The event-log query returned Microsoft-Windows-Sysmon/Operational with IsEnabled = True.

Assessment: Service and log availability were confirmed. These checks do not establish that every desired event type is configured.

Finding 2 — Notepad process creation

Sysmon Event ID 1 recorded Notepad launched with a command line containing SOC_Hunt_Test. The event included the process image, process ID, parent image, user context, integrity level, and a SHA-256 hash.

Assessment: This was intentionally generated benign activity and demonstrated that process-creation details could be retrieved.

Finding 3 — PowerShell process creation

A Sysmon Event ID 1 record showed C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe with command line -NoProfile -Command "Get-Date". The recorded parent image was PowerShell, with parent process ID 7616. The event time was 15 October 2026, 14:14:59 local time as displayed by PowerShell. A related conhost.exe event was also observed.

Assessment: This confirmed PowerShell process-creation telemetry and command-line information. PowerShell spawning another PowerShell process is not inherently malicious; context is required.

Finding 4 — Recent process baseline review

Recent records shown in the review included Microsoft Edge, Zoom, dllhost.exe, conhost.exe, sdbinst.exe, and other Windows activity. The Zoom event contained --action=UpdateSchedule, which appears consistent with update scheduling, but the screenshot alone does not independently verify the binary or its legitimacy.

Assessment: The visible sample appeared consistent with routine application or operating-system activity. No malicious activity was established from the visible records. This was a limited baseline review, not a comprehensive threat hunt.

Limitations

- The review focused primarily on Sysmon Event ID 1.
- Only a limited number of recent events were reviewed.
- No dedicated Sysmon configuration file or broader detection rule set was validated.
- No network, persistence, file modification, registry, authentication, or PowerShell Script Block Logging investigation was completed.
- A benign-looking process name or command line does not prove legitimacy; signature, hash reputation, parent process, timing, and surrounding events may be needed.
- Screenshots may contain a Windows username or device name. Redact these details in public copies.

Recommendations

9. Save the numbered screenshots in evidence/ and verify that each shows the relevant command and output.
10. Export a small, relevant set of Sysmon events to logs/ after sanitising personal identifiers.
11. Document the exact PowerShell queries in rules/.
12. Add a controlled hunt for suspicious command-line patterns and validate leads against parent process, user, timestamp, and file reputation.
13. Consider a reviewed Sysmon configuration and separately assess Windows PowerShell logging.
14. Do not claim a clean bill of health from a limited event sample.

8. Evidence index

Suggested filename	Evidence
01-project-structure.png	Project folders, if captured
02-administrator-powershell.png	Administrator PowerShell, if captured
03-sysmon-executable.png	Executable details, if captured
04-sysmon-installation.png	Successful installation output
05-sysmon-verification.png	Running service and enabled event log
06-notepad-process-event.png	Notepad Event ID 1
08-powershell-process-events.png	PowerShell test event
09-process-parent-investigation.png	Process and parent-process fields

10-powershell-hunting-query.png	Filtered PowerShell query
11-recent-process-review.png	Recent process review

The filenames above are a suggested index. Only include files that have actually been saved.

Lessons learned

Sysmon is a telemetry source, not a verdict. Installing Sysmon and confirming its log is enabled are only the first steps. An analyst must inspect relevant events and confirm that the expected telemetry is being collected.

Event ID 1 helps reconstruct process execution. Process-creation events can show the executable path, command line, process ID, parent process, user context, and available hashes. These fields help establish what ran and how it was launched.

Command-line context matters. The harmless Get-Date test showed that command-line arguments can help explain a process's purpose. A command or tool name alone is not enough to classify activity as malicious.

Parent-child relationships support investigation. A process's parent can provide useful context, but familiar or unusual parent-child combinations should be assessed with timestamps, user context, paths, and surrounding events.

Benign test activity is useful for validation. Using Notepad and Get-Date created controlled events that could be searched without executing malware. This helped validate the collection and query workflow.

A search with no results is not automatically a failure. A query may return no output because the event is outside the selected sample, the filter is too narrow, or the expected field differs. Validate the search method and widen the scope carefully.

Do not overstate findings. The reviewed records did not establish malicious activity. A limited sample cannot prove an endpoint is clean, and an apparently ordinary process should not be declared legitimate without sufficient context.

Evidence handling is part of the work. Screenshots and notes should show the relevant command and output, use clear filenames, and be sanitised before public release to avoid exposing usernames or device names.

Conclusion

This lab demonstrated Sysmon installation and basic validation, retrieval of Event ID 1 records, and analysis of process command lines and parent-child relationships. Controlled Notepad and PowerShell tests produced useful telemetry for a beginner endpoint investigation workflow. The work establishes a foundation for more advanced hunting, but it does not establish that the system is free of threats or that a real malicious incident was detected.